

Finite-Blocklength Latency Minimization for Uplink and Downlink Communications: Method Results: Latency

July 6, 2026

Abstract

This document presents the latency results for uplink and downlink using the same terminology as the companion theory report. Throughout the discussion, the *Training Only Method* is used as the benchmark for the *Training + Testing Method*. The main outcome is different across the two links. In the downlink, the benchmark remains clearly stronger, especially in payload completion, where final total latency is 0.0358 s for the *Training Only Method* and 0.0735 s for the *Training + Testing Method*. In the uplink, the gap is much smaller, with final total latency 0.024667 s for the *Training Only Method* and 0.025200 s for the *Training + Testing Method*.

1 Scope and Naming

This document is latency-only. The terminology is aligned with the companion theory report:

- **Training Only Method:** direct online optimization during scheduling,
- **Training + Testing Method:** a separate training stage followed by testing-stage inference,
- **Payload completion:** the scenario where each user continues until its remaining payload is exhausted.

The *Training Only Method* is treated as the benchmark, and the *Training + Testing Method* is evaluated against it.

2 Downlink

The downlink results show the clearest gap between the benchmark and the *Training + Testing Method*. Table 1 summarizes the scenario-level totals.

Table 1: Downlink latency summary

Scenario	Method	Init. total (s)	Final total (s)	Final avg (s)	Red. (%)
Payload completion	Training Only	0.3050	0.0358	0.01193	88.26
Payload completion	Training + Testing	0.3050	0.0735	0.02451	75.89

2.1 Payload Completion Results and Comparison

Downlink payload completion is the strongest benchmark test. The *Training Only Method* reduces total latency from 0.3050 s to 0.0358 s, while the *Training + Testing Method* reaches 0.073533 s. The testing-stage method therefore improves substantially, but it does not yet match the benchmark.

The per-user pattern in Figure 1 clarifies where that gap comes from. User 1 is already close to the benchmark, with 0.009333 s versus 0.010933 s. The main shortfall is on users 0 and 2. For user 0, the final latency increases from 0.013200 s at the benchmark to 0.028467 s for the *Training + Testing Method*. For user 2, it increases from 0.013267 s to 0.034133 s. The main limitation is therefore the slower-user tail.

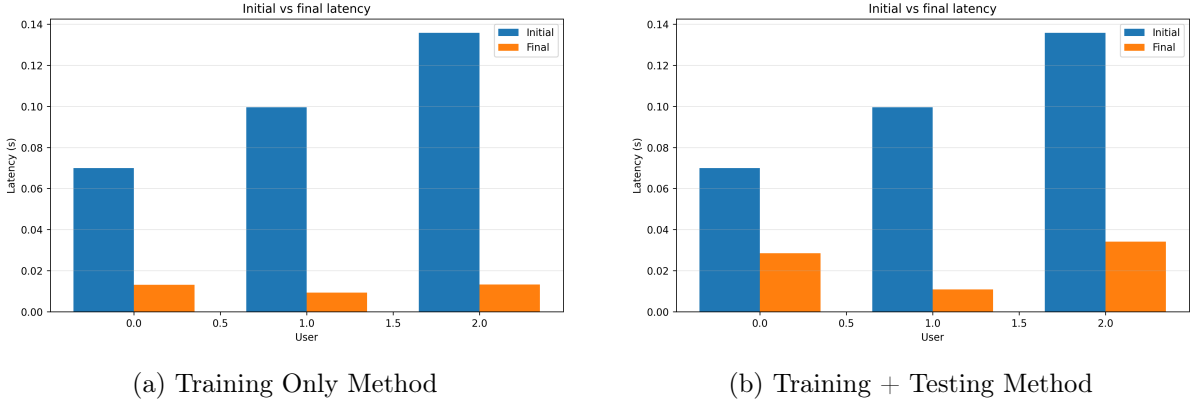


Figure 1: Downlink payload-completion latency comparison. The *Training Only Method* compresses all three user latencies strongly, while the *Training + Testing Method* leaves a visibly longer tail on users 0 and 2.

3 Uplink

In the uplink, the *Training + Testing Method* stays much closer to the benchmark. Table 2 shows that the payload gap remains small.

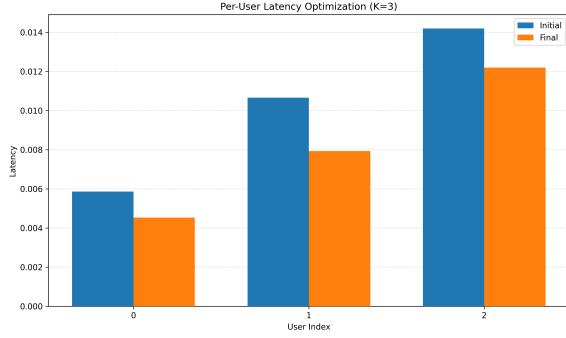
Table 2: Uplink latency summary

Scenario	Method	Init. total (s)	Final total (s)	Final avg (s)	Red. (%)
Payload completion	Training Only	0.0307	0.0247	0.00822	19.74
Payload completion	Training + Testing	0.0307	0.0252	0.00840	18.00

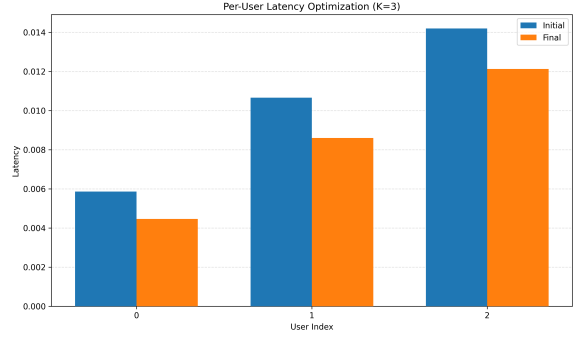
3.1 Payload Completion Results and Comparison

In uplink payload completion, the benchmark and the *Training + Testing Method* are nearly tied. Final total latency is 0.024667 s for the benchmark and 0.025200 s for the *Training + Testing Method*.

Figure 2 confirms this close match. User 0 finishes very slightly better under the *Training + Testing Method* (0.004467 s) than under the *Training Only Method* (0.004533 s). User 2 also ends almost identically (0.012133 s versus 0.012200 s). The main visible difference is user 1, where the *Training Only Method* reaches 0.007933 s instead of 0.008600 s. This user-1 gap is what gives the benchmark the better total latency, but the margin remains small.



(a) Training Only Method



(b) Training + Testing Method

Figure 2: Uplink payload-completion latency comparison. Both methods shorten all three users and end with nearly the same per-user latency profile.

4 Asynchronality Results

The asynchronality results are summarized across both links in Tables 3 and 4. The benchmark remains the clearest reference. Its strongest synchronization gain appears in downlink payload completion, while the uplink results are much closer across the two methods.

Table 3: Downlink asynchronality summary

Scenario	Method	Init. async. (s)	Final async. (s)	Red. (%)
Payload completion	Training Only	0.131600	0.007867	94.02
Payload completion	Training + Testing	0.131600	0.046400	64.74

Table 4: Uplink asynchronality summary

Scenario	Method	Init. async. (s)	Final async. (s)	Red. (%)
Payload completion	Training Only	0.016667	0.015333	8.00
Payload completion	Training + Testing	0.016667	0.015333	8.00

4.1 Downlink

Downlink payload completion is the clearest synchronization win for the *Training Only Method*. Its asynchronality sum falls to 0.007867 s, compared with 0.046400 s for the *Training + Testing Method*. The gap remains visible in the figure pair, where the benchmark finishes with a much tighter final spread.

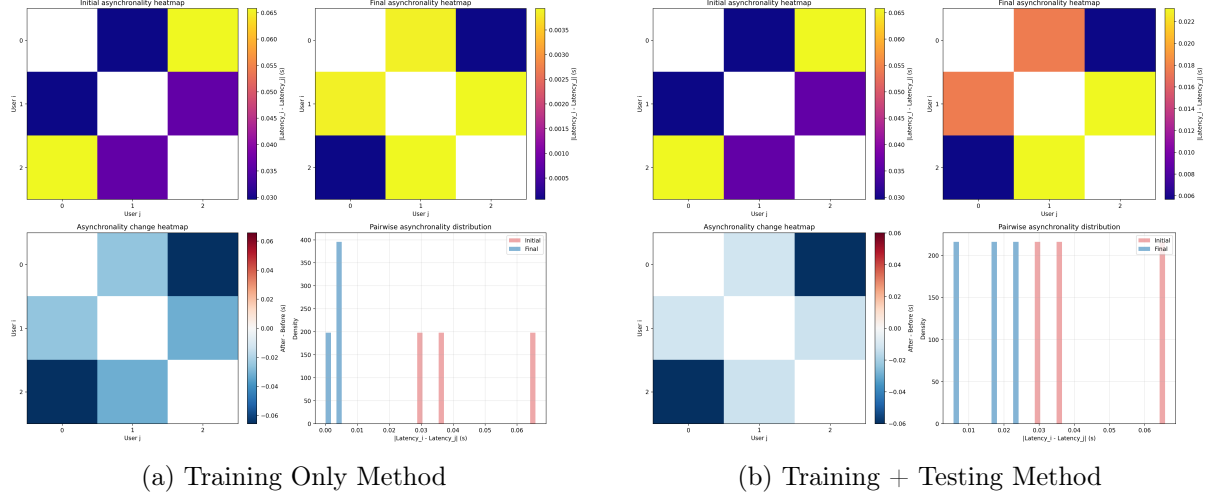


Figure 3: Downlink payload-completion asynchronality comparison. The benchmark produces the tightest final alignment, while the *Training + Testing Method* still leaves a wider residual delay spread.

4.2 Uplink

The uplink results are much tighter across methods. In payload completion, both methods start from 0.016667 s and finish at 0.015333 s, so the final synchronization outcome is effectively the same. This matches the small latency gap reported earlier and suggests that, in this case, the testing-stage method preserves the benchmark synchronization pattern well.

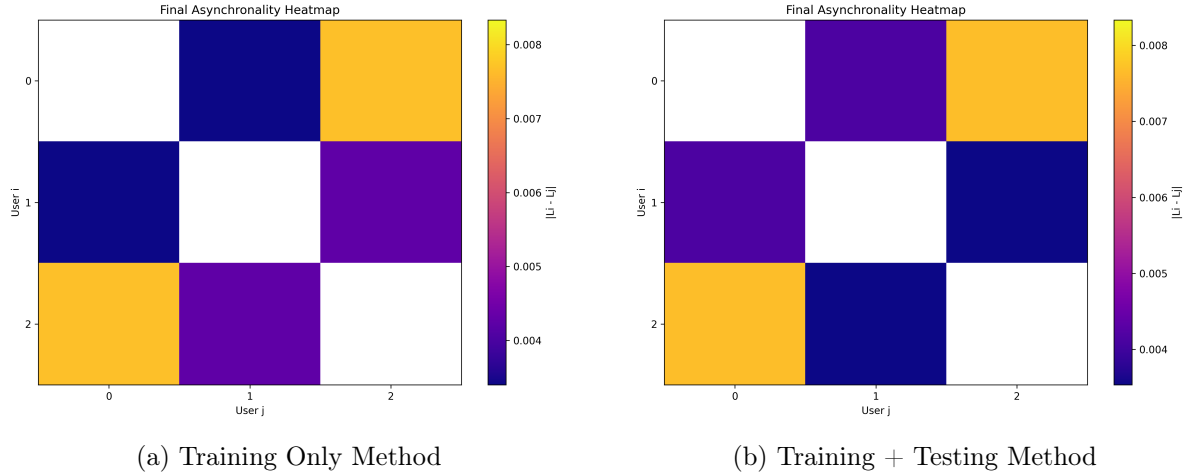


Figure 4: Uplink payload-completion asynchronality comparison. Both methods end with essentially the same final synchronization pattern.

5 Latency Takeaways

The latency results can be read benchmark-first. In the downlink, the *Training + Testing Method* still trails the *Training Only Method*, and the gap is most visible in payload completion, where the slower-user latency tail remains much longer.

In the uplink, the picture is much more encouraging. For payload completion, the *Training + Testing Method* stays close to the benchmark, with only a small total-latency gap. The main takeaway is therefore not a symmetric head-to-head framing, but a benchmark assessment: the

Training + Testing Method is already close in the uplink, while the downlink remains the main area where further latency improvement is needed.